

Alejandro Batres

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Professional Summary

Result driven Data Engineer with 3+ years of experience designing and maintaining scalable data platforms. Proficient in Python, SQL, and data technologies, primarily focused on building high quality, scalable automated solutions. Passionate about utilizing modern technology tools to improve automation and leveraging data for maximum business impact.

TransUnion

Data Engineer - Automation

White Plains, NY

Dec 2024 - Present

- Developed a Redshift and PostgreSQL-based data validation framework utilizing SQL and Python to perform comprehensive data quality checks through an automated workflow, reducing manual effort by 90% and decreasing process and validation time from 1 week to 1 day.
- Built and automated a data retrieval process for UK GDPR (United Kingdom General Data Protection Regulation) compliance by leveraging Python packages psycogp2 and boto3 to identify and retrieve account data based on subject access requests, generate a PDF report detailing the data used by the company, and delete or suppress the account as per the subject's decision. Ensured that the company adhered to GDPR regulations, improved data privacy measures, and reduced the risk of non-compliance penalties by automating the handling of subject access requests.
- Implemented scalable backend services for enrolling clients across multiple regions into a proprietary Performance Insights platform, leveraging AWS EC2 and Airflow to streamline the enrollment process and reduce onboarding time of new clients from two weeks to three days.
- Redesigned malicious-intent reporting by delivering a fully-automated, Airflow-orchestrated Python pipeline using AWS analytics services, built-in data quality checks, and failure notifications. Eliminating manual execution, reducing manual validation by 50%, and ensuring consistent quarterly delivery.
- Delivered client-facing data quality visibility by developing a front-end application that enables users to view data quality checks given their testing preferences and exports client-ready Excel reports utilizing Python packages PyQt and openpyxl, reducing manual effort by 95%.

Data Engineer - Cloud Operations

Dec 2022 - Dec 2024

- Designed and built a serverless Python (boto3) application to automate BAU data movement from AWS Redshift to S3 Data Lake and catalog datasets in AWS Glue with detailed logging, replacing Glue Crawlers to catalog data and reducing costs by 70 percent.
- Developed a core data migration pipeline to transfer SQL Server data to AWS S3 utilizing Postgres, airflow, PySpark, and Python, enabling scalable data categorization and storage workflows, processing about 500GB of data daily.
- Contributed to an AWS Cloud Migration initiative for migrating about 4PB of on-premise data to the cloud using AWS S3, EC2, Redshift, and MSSQL, supporting large-scale data platform modernization.

Allevia

Software Engineering Intern

Remote

Aug 2021 - Jan 2022

- Designed an optimized relational model and data pipeline for the collection and modeling of user input utilizing AWS S3 for secured storage of user data and AWS Lambda for the processing of user metrics.
- Implemented an interactive user-faced mobile application using the React Native for deployment on both IOS and Android with a concentration on accessibility and ease of use.

Skills

Languages: Python, SQL, C++, Typescript

Frameworks: Pandas, Numpy, Airflow, Psycogp2, SQLAlchemy, Boto3 API, PyTorch, React

Cloud Technologies: Airflow, AWS Services (Redshift, Lambda, EC2, S3, Glue), PostgreSQL, Gitlab CI/CD

Education

New York University, Tandon School of Engineering

Bachelor of Science in Mathematics; Minor in Computer Science

Brooklyn, NY

Aug 2018 - May 2022